Variation in patterns of selection and nonlinear selection

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Spatiotemporal variation in phenotypic selection

Environmental variation is expected to generate variation in patterns of selection over time and space. For instance, individuals from different populations may experience varying selective environments and pressures, such as different pollinators, herbivores and predators. These fluctuating selective environments can lead to variation in individuals trait values and in the individuals fitness, ultimately altering the relationship between phenotypic traits and fitness (i.e. selection). The opportunity for selection, I (variance in relative fitness $I = var(\frac{W}{\bar{W}})$) may thus also vary across different environments.

Similarly, the abiotic environment (nutrient availability, soil moisture, temperature) may also vary in space and time, even within the same population, potentially altering the strength and mode of selection. Thus, in order to understand how fluctuations in selection may influence biodiversity, we need to address the spatial and temporal scales at which selection varies (Albertsen et al. 2021).

Typically, selection studies addressing spatiotemporal variation in selection have followed the regression approach by Lande and Arnold (1983) to estimate phenotypic linear selection gradients β for each trait in each environment (e.g. population, year, experimental treatment) followed by analysis of covariance (ANCOVA) to test for differences in mean selection gradients β among different environments.

Recent approaches, as the one proposed by Albertsen et al. (2021), now allow to comprehensively address variation in phenotypic selection over time and space by accounting for the uncertainty associated to any estimate of phenotypic selection and thus quantifying the magnitude of variation in selection.

Here, we will use a dataset from an selection study on different experimental populations (treatments) of the plant common morning glory, to address variation in phenotypic selection.

Exercise 1: Selection on *Ipomoea purpurea* floral traits under water deficit

In this study by Garcia et al. (2023), we set up a common garden with the common morning glory *Ipomoea purpurea*. We studied phenotypic selection in three different experimental populations or treatments consisting on 1) well-watered plants (watered every other day) exposed to open pollination and which we called *control* treatment, 2) plants under water deficit named *drought* treatment, and 3) well-watered plants with restricted access to pollinators, called *pollinator restriction* treatment. The motivation of this study was to address whether two selective agents on floral traits such as pollinators and abiotic resources (water availability) that are expected to change within a context of global change (pollinator decline and increasing droughts) may have an impact on phenotypic selection on floral traits related to plant pollination.

During the flowering season, we measured several phenotypic traits that function as floral signals and rewards to pollinators including flower size, nectar volume and concentration, and plant size (estimated from the diameter of the plant's main stem). We counted the number of fruits and estimated seed set (mean number of seeds per fruit multiplied by fruits) as a reproductive fitness measure.

```r
ipo= read.csv("ipomoea_data.csv")
head(ipo)
```

```
##   Plant_ID Block Source Treatment date_sampling days_setup nectar_vl
## 1      BN1     1     BN         C    2021-08-13          1  2.589844
## 2     BN10     4     BN         C    2021-09-01         20  2.291406
## 3     BN12     1     BN         C    2021-08-14          2  1.092188
## 4     BN13     1     BN         C    2021-08-13          1  2.307031
## 5     BN17     4     BN         C    2021-09-02         21  2.528125
## 6     BN19     1     BN         C    2021-08-13          1  1.374219
##   nectar_conc fl_width fl_lenght mean_size diameter buds flowers buds_2
## 1       42.35   44.215    46.435  45.31141     4.69    0       1      0
## 2       39.65   35.605    45.795  40.37983     4.84    0       0      0
## 3       38.40   36.320    38.285  37.28956     5.16    0       0      0
## 4       36.75   41.250    43.510  42.36493     5.22    0       0      0
## 5       41.65   57.875    53.640  55.71728       NA   NA      NA     NA
## 6       40.90   37.950    40.305  39.10978     4.83    2       5      0
##   flowers_2 init_fruits init_fruits_2 mature_fruits mature_fruits_2 tot_fruits
## 1         0          45            38            87              90        132
## 2         0          32            20            93              93        125
## 3         0         106            76            20              34        126
## 4         0          27            16            79              79        106
## 5        NA          NA            NA            NA              NA         NA
## 6         1          68            52            13              25         81
##   tot_fruits2 fruit_tot mean_seeds seedset
## 1         128     130.0        5.3   678.4
## 2         113     119.0        5.5   621.5
## 3         110     118.0        4.6   506.0
## 4          95     100.5        4.9   465.5
## 5          NA        NA        3.5      NA
## 6          77      79.0        4.8   369.6
```

```r
ipo$Treatment= as.factor(ipo$Treatment)
ipo[1:5, 7:12]# the phenotypic traits
```

```
##   nectar_vl nectar_conc fl_width fl_lenght mean_size diameter
## 1  2.589844       42.35   44.215    46.435  45.31141     4.69
## 2  2.291406       39.65   35.605    45.795  40.37983     4.84
## 3  1.092188       38.40   36.320    38.285  37.28956     5.16
## 4  2.307031       36.75   41.250    43.510  42.36493     5.22
## 5  2.528125       41.65   57.875    53.640  55.71728       NA
```

#Phenotypic correlations between pairs of traits

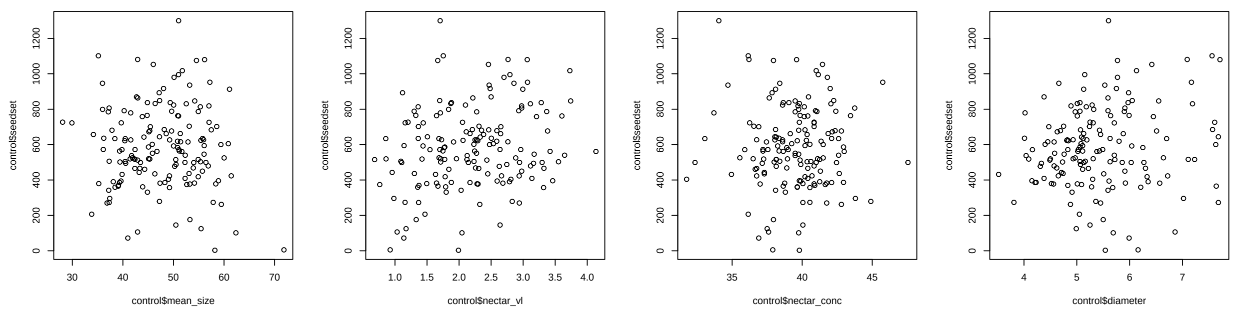
```r
cor.ipo= cor(ipo[,c(7:10, 12)], use="pairwise")
signif(cor.ipo, 2)
```

```
##             nectar_vl nectar_conc fl_width fl_lenght diameter
## nectar_vl       1.000       0.083    0.430    0.4500   0.0870
## nectar_conc     0.083       1.000    0.120    0.0780   0.0220
## fl_width        0.430       0.120    1.000    0.7800   0.0270
## fl_lenght       0.450       0.078    0.780    1.0000  -0.0034
## diameter        0.087       0.022    0.027   -0.0034   1.0000
```

#Explore the data in each experimental population (treatment)

```
#In the control treatment (well-watered plants)
control= ipo %>%
  filter(Treatment %in% "C")

par(mfcol= c(1,4))
plot(control$mean_size, control$seedset)
plot(control$nectar_vl, control$seedset)
plot(control$nectar_conc, control$seedset)
plot(control$diameter, control$seedset)
```

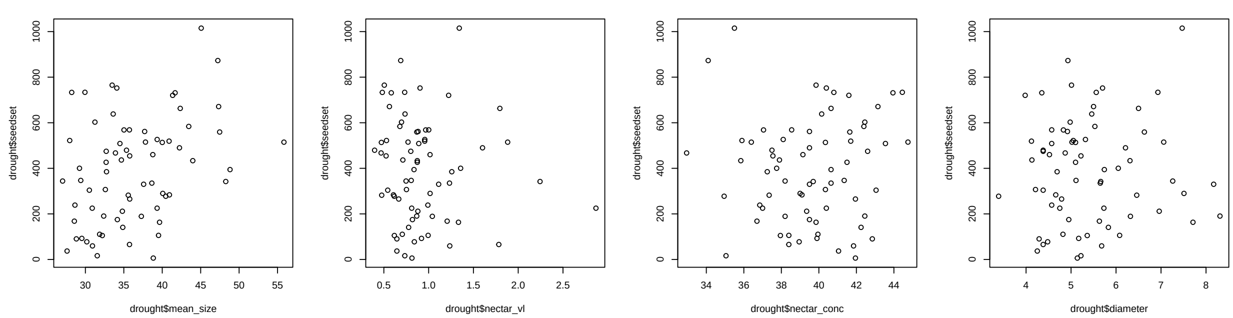



```
#Estimating relative fitness in the control treatment
control$relfit= control$seedset/mean(control$seedset, na.rm= TRUE)
```

Notice that we scale the traits for each population/treatment/year separately. In the same way as we do for trait standardization, we estimate the relative fitness $w = W/\bar{W}$ for each population, this is dividing individual fitness by its population's mean fitness.

```
#In the drought treatment (plants in water deficit)
drought= ipo %>%
  filter(Treatment %in% "D")

par(mfcol= c(1,4))
plot(drought$mean_size, drought$seedset)
plot(drought$nectar_vl, drought$seedset)
plot(drought$nectar_conc, drought$seedset)
plot(drought$diameter, drought$seedset)
```

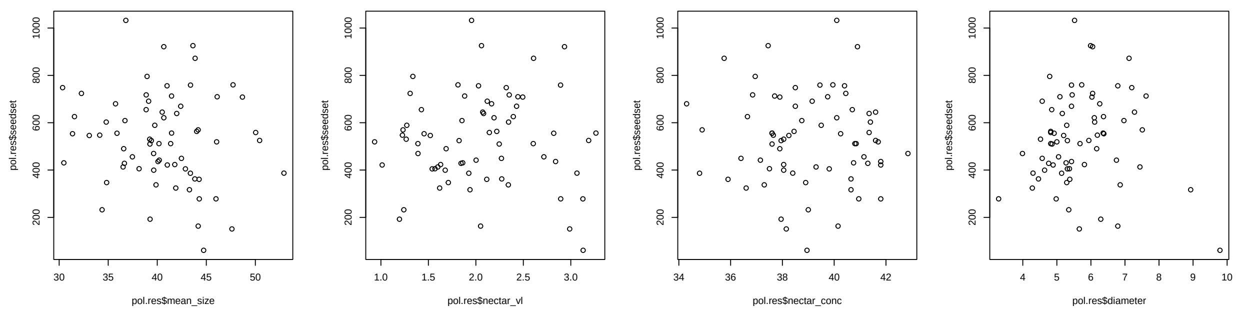


```r
#Estimating relative fitness in the drought (water deficit) treatment
drought$relfit= drought$seedset/mean(drought$seedset, na.rm= TRUE)
```

#In the pollinator restriction treatment (well-watered plants without pollinator access)

```r
#In the control treatment (well-watered plants)
pol.res= ipo %>%
  filter(Treatment %in% "PR")

par(mfcol= c(1,4))
plot(pol.res$mean_size, pol.res$seedset)
plot(pol.res$nectar_vl, pol.res$seedset)
plot(pol.res$nectar_conc, pol.res$seedset)
plot(pol.res$diameter, pol.res$seedset)
```




```r
#Estimating relative fitness in the pollination restriction treatment
pol.res$relfit= pol.res$seedset/mean(pol.res$seedset, na.rm= TRUE)
```

#Opportunity for selection (variance in relative fitness) in each treatment

```r
#In the control treatment
IC = var(control$relfit, na.rm= TRUE)
IC
```

```
## [1] 0.157721
```

```r
#In the drought treatment
ID = var(drought$relfit, na.rm= TRUE)
ID
```

```
## [1] 0.3409269
```

```r
#In the pollinator restriction treatment
IPR = var(pol.res$relfit, na.rm= TRUE)
IPR
```

```
## [1] 0.131094
```

```r
#Opportunity for selection is higher in the water deficit experimental population
```

#Multivariate variance-standardized selection gradients:

```r
#In the control treatment
m = lm(relfit~ scale(nectar_vl) + scale(nectar_conc)
+ scale(mean_size) + scale(diameter), data= control, na=na.exclude)
summary(m)$coef
```

```
##                       Estimate Std. Error    t value     Pr(>|t|)
## (Intercept)         1.00085029 0.03096119 32.3259627 1.766101e-69
## scale(nectar_vl)    0.09859594 0.03210311  3.0712270 2.531822e-03
## scale(nectar_conc) -0.02682126 0.03224230 -0.8318657 4.068071e-01
## scale(mean_size)   -0.03589359 0.03273290 -1.0965600 2.745911e-01
## scale(diameter)     0.05844165 0.03150725  1.8548637 6.557876e-02
```

We notice that there is linear selection to increase nectar volume (positive sign of the parameter's Estimate) in flowers in control conditions. This is, there is an increase of 9% (nectar vol Estimate= 0.09) in plant fitness (seed set) per standard deviation increase in nectar volume. In addition, we detect selection to increase plant size (stem diameter), this is, larger plants produce more seeds. In this case seed set increases by 6% per standard deviation increase in plant size (stem diameter Estimate= 0.058)

#Multivariate mean-standardized selection gradients:

```r
#We center the traits to their mean (subtract the mean)
#and mean-scale the traits (divide by the mean)
control$nectar_vl_m =
  (control$nectar_vl-mean(control$nectar_vl, na.rm=T))/mean(control$nectar_vl, na.rm=T)
control$nectar_conc_m =
  (control$nectar_conc-mean(control$nectar_conc, na.rm=T))/mean(control$nectar_conc, na.rm=T)
control$mean_size_m =
  (control$mean_size-mean(control$mean_size, na.rm=T))/mean(control$mean_size, na.rm=T)
control$diameter_m =
  (control$diameter-mean(control$diameter, na.rm = T))/mean(control$diameter, na.rm = T)
```

```r
#Then we estimate the mean-scaled multivariate linear selection gradients:
m.mean = lm(relfit~ nectar_vl_m + nectar_conc_m
+ mean_size_m + diameter_m, data= control, na=na.exclude)

summary(m.mean)$coef
```

```
##                 Estimate Std. Error    t value     Pr(>|t|)
## (Intercept)    1.0008503 0.03096119 32.3259627 1.766101e-69
## nectar_vl_m    0.2940615 0.09574722  3.0712270 2.531822e-03
## nectar_conc_m -0.4356811 0.52373976 -0.8318657 4.068071e-01
## mean_size_m   -0.2195454 0.20021282 -1.0965600 2.745911e-01
## diameter_m     0.3523579 0.18996430  1.8548637 6.557876e-02
```

We detect selection on the same traits, when we mean-scale the traits, but in this case the mean-scaled selection gradients, β_{μ}, for nectar volume and stem diameter (0.294 vs 0.352) rank differently than variance-scaled gradients, β_{σ} (0.098 vs 0.058)

#In the water deficit treatment:

```r
#Variance-scaled multivariate selection gradients:
md = lm(relfit~ scale(nectar_vl) + scale(nectar_conc)
+ scale(mean_size) + scale(diameter), data= drought, na=na.exclude)

summary(md)$coef
```

```
##                         Estimate Std. Error    t value     Pr(>|t|)
## (Intercept)          1.002805276 0.06560678 15.2850857 4.873794e-23
## scale(nectar_vl)    -0.133208586 0.06940373 -1.9193289 5.940343e-02
## scale(nectar_conc)   0.029288669 0.06453968  0.4538087 6.515017e-01
## scale(mean_size)     0.250715982 0.06865726  3.6517041 5.263122e-04
## scale(diameter)      0.006141644 0.06733941  0.0912043 9.276151e-01
```

In contrast to control plants, we observe selection favoring plants with lower nectar volumes (negative sign of the parameter's Estimate) suggesting a cost of nectar rewards in stressful conditions (water deficit). Specifically, fitness decreases by 13% (-0.133) per standard deviation increase in nectar volume. Interestingly, we find strong positive selection on floral size under water deficit (0.250), with an increase of 25% in fitness by a standard deviation increase in flower size.

#Mean-standardized selection gradients in water deficit:

```r
#First we center to the mean (subtract) and mean-scale (divide) the traits:
drought$nectar_vl_m =
  (drought$nectar_vl-mean(drought$nectar_vl, na.rm=T))/mean(drought$nectar_vl, na.rm=T)
drought$nectar_conc_m =
  (drought$nectar_conc-mean(drought$nectar_conc, na.rm=T))/mean(drought$nectar_conc, na.rm=T)
drought$mean_size_m =
  (drought$mean_size-mean(drought$mean_size, na.rm=T))/mean(drought$mean_size, na.rm=T)
drought$diameter_m =
  (drought$diameter-mean(drought$diameter, na.rm=T)) /mean(drought$diameter, na.rm = T)

#Then we estimate the multivariate linear selection gradients:
md.mean = lm(relfit~ nectar_vl_m + nectar_conc_m
+ mean_size_m + diameter_m, data= drought, na=na.exclude)

summary(md.mean)$coef
```

```
##                  Estimate Std. Error    t value     Pr(>|t|)
## (Intercept)     1.0028053 0.06560678 15.2850857 4.873794e-23
## nectar_vl_m    -0.2939973 0.15317715 -1.9193289 5.940343e-02
## nectar_conc_m   0.4655346 1.02583872  0.4538087 6.515017e-01
## mean_size_m     1.5496989 0.42437691  3.6517041 5.263122e-04
## diameter_m      0.0312788 0.34295316  0.0912043 9.276151e-01
```

Again, we observed strong selection on flower size, which is stronger than selection on fitness as a trait($\beta_\mu = 1.549 > 1$).

As we have seen, the direction in patterns of selection on nectar volume differ between treatments, with positive directional selection in plants from control conditions contrasting with negative directional selection in plants under water deficit. Selection to increase flower size in water deficit also differs from what we observed in well-watered conditions (control and PR treatments), where we did not detect selection acting on flower size.

To estimate whether selection varies between treatments, we perform analysis of covariance (ANCOVA) in which we compare the means of the selection gradients β between treatments. For this, we need to add the interaction terms between each trait and the variable that represents the different treatments (here "Treatment").

```
#Estimate relative fitness within each treatment:
ipo.treat=ipo %>%
  group_by(Treatment)%>%
  mutate("relfit"= seedset/mean(seedset, na.rm= TRUE))%>%
  ungroup()
```

Here, because of the unbalanced design (large difference in sample size between treatments, with larger size in the control treatment) we performed a paired ANCOVA between the control and the two manipulative treatments (i.e., water deficit and and poll. restriction).

Control vs water deficit:

```
#Fit the ANCOVA model to test for differences in selection between
#control and water deficit:
ipo.treat.cd = ipo.treat %>%
  filter(!Treatment %in% "PR") %>% #exclude pol.res treatment for comparison
  droplevels()

#Notice that the ANCOVA includes the interaction (*) between each trait and the treatment variable
m.ancova.cd= aov(relfit ~ scale(nectar_vl)+ scale(nectar_conc)
+ scale(mean_size)+ scale(diameter)+ scale(nectar_vl)*Treatment+
scale(nectar_conc)*Treatment+ scale(mean_size)*Treatment+
  scale(diameter)*Treatment, na= na.exclude, data = ipo.treat.cd)
summary(m.ancova.cd)
```

```
##                               Df Sum Sq Mean Sq F value  Pr(>F)
## scale(nectar_vl)               1   0.44   0.444   2.304  0.1306
## scale(nectar_conc)             1   0.00   0.000   0.000  0.9920
## scale(mean_size)               1   0.03   0.025   0.130  0.7185
## scale(diameter)                1   0.39   0.388   2.010  0.1577
## Treatment                      1   0.48   0.482   2.498  0.1155
## scale(nectar_vl):Treatment     1   0.76   0.761   3.943  0.0484 *
## scale(nectar_conc):Treatment   1   0.15   0.147   0.762  0.3836
## scale(mean_size):Treatment     1   3.79   3.786  19.628 1.5e-05 ***
## scale(diameter):Treatment      1   0.16   0.158   0.821  0.3659
## Residuals                    214  41.28   0.193
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 12 observations deleted due to missingness
```

The ANCOVA indicates that phenotypic selection on nectar volume and floral size differs between the control and the water deficit treatment.

Control vs poll. restriction:

```
ipo.treat.cpr = ipo.treat %>%
  filter(!Treatment %in% "D") %>% #exclude water deficit treatment for #comparison
  droplevels()

m.ancova.cpr= aov(relfit ~ scale(nectar_vl)+ scale(nectar_conc)+
scale(mean_size)+ scale(diameter)+ scale(nectar_vl)*Treatment+
  scale(nectar_conc)*Treatment+ scale(mean_size)*Treatment+ scale(diameter)*Treatment,
na= na.exclude, data = ipo.treat.cpr)

summary(m.ancova.cpr)
```

```
##                                Df Sum Sq Mean Sq F value Pr(>F)
## scale(nectar_vl)                1  0.950  0.9503   6.561 0.0111 *
## scale(nectar_conc)              1  0.093  0.0927   0.640 0.4246
## scale(mean_size)                1  0.306  0.3058   2.111 0.1477
## scale(diameter)                 1  0.262  0.2618   1.808 0.1802
## Treatment                       1  0.053  0.0531   0.366 0.5456
## scale(nectar_vl):Treatment      1  0.273  0.2732   1.887 0.1710
## scale(nectar_conc):Treatment    1  0.004  0.0044   0.030 0.8626
## scale(mean_size):Treatment      1  0.040  0.0395   0.273 0.6018
## scale(diameter):Treatment       1  0.229  0.2294   1.584 0.2096
## Residuals                     214 30.993  0.1448
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 9 observations deleted due to missingness
```

In contrast, selection did not differ between well-watered plants with varying access to pollinators (i.e control vs. pollinator restriction treatment)

#ANCOVA on mean-standardized multivariate selection gradients:

```
#Another way to center and scale the traits by their mean within each treatment
ipo.treat= ipo.treat %>%
  group_by(Treatment)%>%
  mutate(across(c(nectar_vl, nectar_conc, mean_size, diameter),
              ~(.x-mean(.x, na.rm = TRUE))/mean(.x, na.rm = TRUE),
              .names= "{col}_m"))%>%
  ungroup()

#Fit the ANCOVA model to test for differences in selection between control and water deficit:
ipo.treat.cd= ipo.treat %>%
  filter(!Treatment %in% "PR") %>% #exclude pol.res treatment for
  #comparison
  droplevels()

m.ancova.cdm= aov(relfit ~ nectar_vl_m+ nectar_conc_m+ mean_size_m+
diameter_m+ nectar_vl_m*Treatment+ nectar_conc_m*Treatment+
  mean_size_m*Treatment+ diameter_m*Treatment,
na= na.exclude, data = ipo.treat.cd)

summary(m.ancova.cdm)
```

```
##                          Df  Sum Sq  Mean Sq  F value    Pr(>F)    
## nectar_vl_m               1    0.30    0.304    1.578    0.2104    
## nectar_conc_m             1    0.00    0.000    0.000    0.9915    
## mean_size_m               1    0.45    0.455    2.358    0.1262    
## diameter_m                1    0.35    0.355    1.838    0.1766    
## Treatment                 1    0.00    0.000    0.000    0.9838    
## nectar_vl_m:Treatment     1    1.25    1.252    6.488    0.0116 *  
## nectar_conc_m:Treatment   1    0.16    0.162    0.840    0.3606    
## mean_size_m:Treatment     1    3.50    3.503   18.158 3.05e-05 ***
## diameter_m:Treatment      1    0.16    0.161    0.836    0.3615    
## Residuals               214   41.28    0.193                      
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 12 observations deleted due to missingness
```

The ANCOVA approach indicates variation in phenotypic selection between the control and water deficit treatment. To evaluate the magnitude of variation in phenotypic selection gradients between treatments, we perform the Error-corrected standard deviation of the selection gradients (Albertsen et al. 2021).

#Error-corrected variance of the mean-standardized phenotypic selection gradients:

To further evaluate variation in phenotypic selection between treatments, we follow the approach proposed by Albertsen et al (2021) that estimates the mean dispersion of selection gradients.

We start by fitting the multivariate selection model on our set of traits on each treatment separately. We will focus on the comparison between the control and the drought treatment for which we have seen that the ANCOVA test supports variation in patterns of selection.

This time, we will fit the multivariate selection models just on the mean-centered traits (subtracting the trait mean), and later we will estimate the mean-scaled gradients by multiplying the raw model's estimates by the trait means.

```
#Control treatment.
#Mean-center the traits (subtract the mean trait value)
control= control %>%
  mutate(
    nectar_vl_mean_cent   = nectar_vl - mean(nectar_vl, na.rm = TRUE),
    nectar_conc_mean_cent = nectar_conc - mean(nectar_conc, na.rm = TRUE),
    mean_size_mean_cent   = mean_size - mean(mean_size, na.rm = TRUE),
    diameter_mean_cent    = diameter - mean(diameter, na.rm = TRUE))
```

Fit the multivariate regression (selection) model on the mean-centered traits

```
control_mean_mod= lm(relfit~ nectar_vl_mean_cent + nectar_conc_mean_cent
+ mean_size_mean_cent + diameter_mean_cent, data= control, na=na.exclude)
```

Then we calculate the mean and standard variation for each trait

```
control_mean_traits= control %>%
  summarise_at(c("nectar_vl",
                   "nectar_conc",
                   "mean_size",
                   "diameter"), mean,
               na.rm=TRUE)
```

Reshape the format with melt function

```
control_mean_traits= melt(control_mean_traits, value.name = "Trait_Mean", variable.name = "Trait")
control_mean_traits
```

```
##         Trait Trait_Mean
## 1   nectar_vl   2.198698
## 2 nectar_conc  39.472963
## 3   mean_size  47.635168
## 4    diameter   5.464395
```

```
control_sd_traits= control %>%
  summarise_at(c("nectar_vl",
                      "nectar_conc",
                      "mean_size",
                      "diameter"), sd,
               na.rm=TRUE)
control_sd_traits
```

```
##   nectar_vl nectar_conc mean_size  diameter
## 1  0.737202    2.430022  7.787899 0.9063178
```

```
control_sd_traits= melt(control_sd_traits, value.name = "Trait_SD", variable.name = "Trait")
control_sd_traits
```

```
##         Trait  Trait_SD
## 1   nectar_vl 0.7372020
## 2 nectar_conc 2.4300218
## 3   mean_size 7.7878994
## 4    diameter 0.9063178
```

```
control_summary_traits= left_join(control_mean_traits,
                                               control_sd_traits,
                                               by = c("Trait" = "Trait"))
```

Round to four decimals

```
control_summary_traits= control_summary_traits %>%
  mutate_if(is.numeric, round, digits = 4)

control_summary_traits
```

```
##         Trait Trait_Mean Trait_SD
## 1   nectar_vl     2.1987   0.7372
## 2 nectar_conc    39.4730   2.4300
## 3   mean_size    47.6352   7.7879
## 4    diameter     5.4644   0.9063
```

Create a dataframe with the model's information (estimate for each trait, SE and p value).

To get the model's estimates for each trait, we just extract the model coefficients. To estimate the standard error (SE) of the model's estimates (Estimate) for each trait, we calculate the standard errors as the square root (sqrt) of the variance, which is the diagonal of the variance-covariance (vcov(model)) matrix of the model.

We only obtain the estimates, SE and p. values, for the traits (model's coefficients[2:5]) and exclude the intercept (model's coefficient[1])

```
control_betas=
  data.frame(Trait = c("nectar_vl",
                       "nectar_conc",
                       "mean_size",
                       "diameter"),
             Estimate = c(control_mean_mod$coefficients[2],
                          control_mean_mod$coefficients[3],
                          control_mean_mod$coefficients[4],
                          control_mean_mod$coefficients[5]),
             SE_Model = c(sqrt(diag(vcov(control_mean_mod))[2]),
                          sqrt(diag(vcov(control_mean_mod))[3]),
                          sqrt(diag(vcov(control_mean_mod))[4]),
                          sqrt(diag(vcov(control_mean_mod))[5])),
             p = c(summary(control_mean_mod)$coefficients[2, "Pr(>|t|)"],
                   summary(control_mean_mod)$coefficients[3, "Pr(>|t|)"],
                   summary(control_mean_mod)$coefficients[4, "Pr(>|t|)"],
                   summary(control_mean_mod)$coefficients[5, "Pr(>|t|)"]))
control_betas
```

```
##                             Trait     Estimate    SE_Model           p
## nectar_vl_mean_cent     nectar_vl  0.133743456 0.043547240 0.002531822
## nectar_conc_mean_cent nectar_conc -0.011037457 0.013268316 0.406807121
## mean_size_mean_cent     mean_size -0.004608893 0.004203046 0.274591117
## diameter_mean_cent       diameter  0.064482508 0.034764014 0.065578756
```

Round to four decimals

```
control_betas= control_betas %>%
  mutate_if(is.numeric, round, digits = 4)
control_betas
```

```
##                             Trait Estimate SE_Model      p
## nectar_vl_mean_cent     nectar_vl   0.1337   0.0435 0.0025
## nectar_conc_mean_cent nectar_conc  -0.0110   0.0133 0.4068
## mean_size_mean_cent     mean_size  -0.0046   0.0042 0.2746
## diameter_mean_cent       diameter   0.0645   0.0348 0.0656
```

```
control_gradients= left_join(control_betas, control_summary_traits,
                                                       by = c("Trait" = "Trait"))
control_gradients
```

```
##         Trait Estimate SE_Model      p Trait_Mean Trait_SD
## 1   nectar_vl   0.1337   0.0435 0.0025     2.1987   0.7372
## 2 nectar_conc  -0.0110   0.0133 0.4068    39.4730   2.4300
## 3   mean_size  -0.0046   0.0042 0.2746    47.6352   7.7879
## 4    diameter   0.0645   0.0348 0.0656     5.4644   0.9063
```

#Estimate mean-scaled selection gradients (and their SE) for each trait As previously mentioned, we multiply the model raw estimates for each trait (Estimate) by the trait mean in order to estimate the mean-scaled selection gradients.

```
control_gradients=  control_gradients %>%
  mutate(Mean_Selection_Gradient = round(Estimate* Trait_Mean, digits = 4)) %>%
  mutate(Mean_SE_Selection_Gradient = round(SE_Model * Trait_Mean, digits = 4)) %>%
  mutate(Treatment = "control")

control_gradients
```

```
##         Trait Estimate SE_Model      p Trait_Mean Trait_SD
## 1   nectar_vl   0.1337   0.0435 0.0025     2.1987   0.7372
## 2 nectar_conc  -0.0110   0.0133 0.4068    39.4730   2.4300
## 3   mean_size  -0.0046   0.0042 0.2746    47.6352   7.7879
## 4    diameter   0.0645   0.0348 0.0656     5.4644   0.9063
##   Mean_Selection_Gradient Mean_SE_Selection_Gradient Treatment
## 1                  0.2940                     0.0956   control
## 2                 -0.4342                     0.5250   control
## 3                 -0.2191                     0.2001   control
## 4                  0.3525                     0.1902   control
```

Now we follow the same steps for the water deficit treatment:

```
#Drought treatment.
#Mean-center the traits (subtract the mean trait value)
drought= drought %>%
  mutate(
    nectar_vl_mean_cent   = nectar_vl - mean(nectar_vl, na.rm = TRUE),
    nectar_conc_mean_cent = nectar_conc - mean(nectar_conc, na.rm = TRUE),
    mean_size_mean_cent   = mean_size - mean(mean_size, na.rm = TRUE),
    diameter_mean_cent    = diameter - mean(diameter, na.rm = TRUE))
```

Fit the multivariate selection models on the mean-centered traits

```
drought_mean_mod= lm(relfit~ nectar_vl_mean_cent + nectar_conc_mean_cent
+ mean_size_mean_cent + diameter_mean_cent, data= drought, na=na.exclude)
```

Then we calculate the mean and standard variation for each trait

```
#drought treatment
drought_mean_traits= drought %>%
  summarise_at(c("nectar_vl",
                      "nectar_conc",
                      "mean_size",
                      "diameter"), mean,
               na.rm=TRUE)
```

```
drought_mean_traits= melt(drought_mean_traits, value.name = "Trait_Mean", variable.name = "Trait")

drought_mean_traits
```

```
##         Trait Trait_Mean
## 1   nectar_vl  0.9272465
## 2 nectar_conc 39.5802703
## 3   mean_size 36.3732825
## 4    diameter  5.3955714
```

```
drought_sd_traits= drought %>%
  summarise_at(c("nectar_vl",
                     "nectar_conc",
                     "mean_size",
                     "diameter"), sd,
               na.rm=TRUE)
drought_sd_traits
```

```
##   nectar_vl nectar_conc mean_size diameter
## 1 0.4201303    2.490155  5.884603 1.059429
```

```
drought_sd_traits= melt(drought_sd_traits, value.name = "Trait_SD", variable.name = "Trait")
drought_sd_traits
```

```
##         Trait  Trait_SD
## 1   nectar_vl 0.4201303
## 2 nectar_conc 2.4901554
## 3   mean_size 5.8846033
## 4    diameter 1.0594292
```

```
drought_summary_traits= left_join(drought_mean_traits,
                                                drought_sd_traits,
                                                by = c("Trait" = "Trait"))
```

Round to four decimals

```
drought_summary_traits= drought_summary_traits %>%
  mutate_if(is.numeric, round, digits = 4)

drought_summary_traits
```

```
##         Trait Trait_Mean Trait_SD
## 1   nectar_vl     0.9272   0.4201
## 2 nectar_conc    39.5803   2.4902
## 3   mean_size    36.3733   5.8846
## 4    diameter     5.3956   1.0594
```

Create a dataframe with the model information (estimate for each trait, SE and p value)

```
drought_betas=
  data.frame(Trait = c("nectar_vl",
                     "nectar_conc",
                     "mean_size",
                     "diameter"),
             Estimate = c(drought_mean_mod$coefficients[2],
```

```
                          drought_mean_mod$coefficients[3],
                          drought_mean_mod$coefficients[4],
                          drought_mean_mod$coefficients[5]),
             SE_Model = c(sqrt(diag(vcov(drought_mean_mod))[2]),
                          sqrt(diag(vcov(drought_mean_mod))[3]),
                          sqrt(diag(vcov(drought_mean_mod))[4]),
                          sqrt(diag(vcov(drought_mean_mod))[5])),
             p = c(summary(drought_mean_mod)$coefficients[2, "Pr(>|t|)"],
                   summary(drought_mean_mod)$coefficients[3, "Pr(>|t|)"],
                   summary(drought_mean_mod)$coefficients[4, "Pr(>|t|)"],
                   summary(drought_mean_mod)$coefficients[5, "Pr(>|t|)"]))
```

Round to four decimals

```
drought_betas= drought_betas %>%
  mutate_if(is.numeric, round, digits = 4)
drought_betas
```

```
##                              Trait Estimate SE_Model      p
## nectar_vl_mean_cent     nectar_vl  -0.3171   0.1652 0.0594
## nectar_conc_mean_cent nectar_conc   0.0118   0.0259 0.6515
## mean_size_mean_cent     mean_size   0.0426   0.0117 0.0005
## diameter_mean_cent       diameter   0.0058   0.0636 0.9276
```

```
drought_gradients= left_join(drought_betas, drought_summary_traits,
                                                     by = c("Trait" = "Trait"))
```

#Estimate the mean-scaled selection gradients in the drought treatment:

```
drought_gradients=  drought_gradients %>%
  mutate(Mean_Selection_Gradient = round(Estimate* Trait_Mean, digits = 4)) %>%
  mutate(Mean_SE_Selection_Gradient = round(SE_Model * Trait_Mean, digits = 4)) %>%
  mutate(Treatment = "drought")

drought_gradients
```

```
##         Trait Estimate SE_Model      p Trait_Mean Trait_SD
## 1   nectar_vl  -0.3171   0.1652 0.0594     0.9272   0.4201
## 2 nectar_conc   0.0118   0.0259 0.6515    39.5803   2.4902
## 3   mean_size   0.0426   0.0117 0.0005    36.3733   5.8846
## 4    diameter   0.0058   0.0636 0.9276     5.3956   1.0594
##   Mean_Selection_Gradient Mean_SE_Selection_Gradient Treatment
## 1                 -0.2940                     0.1532   drought
## 2                  0.4670                     1.0251   drought
## 3                  1.5495                     0.4256   drought
## 4                  0.0313                     0.3432   drought
```

```
data_gradients= rbind(control_gradients, drought_gradients)
data_gradients
```

```
##          Trait Estimate SE_Model      p Trait_Mean Trait_SD
## 1   nectar_vl   0.1337   0.0435 0.0025     2.1987   0.7372
## 2 nectar_conc  -0.0110   0.0133 0.4068    39.4730   2.4300
## 3   mean_size  -0.0046   0.0042 0.2746    47.6352   7.7879
## 4    diameter   0.0645   0.0348 0.0656     5.4644   0.9063
## 5   nectar_vl  -0.3171   0.1652 0.0594     0.9272   0.4201
## 6 nectar_conc   0.0118   0.0259 0.6515    39.5803   2.4902
## 7   mean_size   0.0426   0.0117 0.0005    36.3733   5.8846
## 8    diameter   0.0058   0.0636 0.9276     5.3956   1.0594
##   Mean_Selection_Gradient Mean_SE_Selection_Gradient Treatment
## 1                  0.2940                     0.0956   control
## 2                 -0.4342                     0.5250   control
## 3                 -0.2191                     0.2001   control
## 4                  0.3525                     0.1902   control
## 5                 -0.2940                     0.1532   drought
## 6                  0.4670                     1.0251   drought
## 7                  1.5495                     0.4256   drought
## 8                  0.0313                     0.3432   drought
```

##Estimate the error-corrected SD of the Mean selection gradients:

```
data_gradients_mean_variance= data_gradients %>%
  group_by(Trait) %>%
  summarise(Variance_Mean_Selection_Gradient = var(Mean_Selection_Gradient),
            Mean_Mean_SE_Selection_Gradient = mean(Mean_SE_Selection_Gradient^2),
            Difference = Variance_Mean_Selection_Gradient - Mean_Mean_SE_Selection_Gradient,
Variance_Corrected_Mean_Selection_Gradient =
sqrt(Variance_Mean_Selection_Gradient - Mean_Mean_SE_Selection_Gradient)) %>%
  mutate_if(is.numeric, round, digits = 4)
data_gradients_mean_variance
```

```
## # A tibble: 4 x 5
##   Trait       Variance_Mean_Selection_Gradient Mean_Mean_SE_Selecti~1 Difference
##   <chr>                                  <dbl>                  <dbl>      <dbl>
## 1 diameter                              0.0516                 0.077     -0.0254
## 2 mean_size                             1.56                   0.111      1.45
## 3 nectar_conc                           0.406                  0.663     -0.257
## 4 nectar_vl                             0.173                  0.0163     0.157
## # i abbreviated name: 1: Mean_Mean_SE_Selection_Gradient
## # i 1 more variable: Variance_Corrected_Mean_Selection_Gradient <dbl>
```

We see that in two out of the four traits (floral size and nectar volume) there is variation in phenotypic selection between the two treatments (control vs drought). The magnitude of the variation in selection on floral size was three times (1.205) the magnitude of variation on nectar volume (0.396). In contrast, there is no detectable variation in selection on plant size and nectar concentration beyond sampling error.

Nonlinear selection

#Stabilizing and disruptive selection

Up to now we have seen different examples of linear, also known as directional, phenotypic selection. This mode of selection acts by changing the mean trait values in the population (e.g. plants with larger flowers or lower nectar volumes are selected). However, there are other modes of selection, where selection acts by changing the phenotypic variance in the population rather than the mean. For instance, selection may favor those individuals with trait values closer to the average phenotype (they will have higher fitness), while individuals with trait values close to the extremes of the distribution will show lower fitness values. This mode of selection is known as *stabilizing selection* and it acts decreasing the phenotypic variance in the population. In contrast to stabilizing selection, selection may favor those individuals with extreme trait values (close to the tails of the trait distribution) acting against average individuals. This case of nonlinear selection is called *disruptive selection* and increases the phenotypic variance in the population.

In both cases, we will need curve-fitting approaches to describe the selection surface (the relationship between the individuals phenotypic trait values and fitness). The regression approach by Lande and Arnold (1983) also allow us to quantify nonlinear phenotypic selection gradients by applying second-order polynomial regression. For this, we add a quadratic term to the regression model of relative fitness on phenotype, which can be written as:

$w = \alpha + \sum_i z_i\beta_i + 1/2\sum_i z_i^2\gamma_i + \epsilon_i$

where w is the relative fitness of the individual, α is an intercept, β's are partial regression coefficients (linear regression terms) and γ's are the quadratic coefficients of the regression model of relative fitness (w) on the individual's trait zi values. The 1/2 factor is used to make the quadratic terms equivalent to second derivatives (Arnold 2003).

The quadratic regression terms γ_i's are used to measure nonlinear selection.γ_i's <0 indicate stabilizing selection while γ_i's >0 indicate disruptive selection. It is important to notice that to obtain the nonlinear selection gradients, we need to double the resulting quadratic regression coefficients, γ_i, as well as their standard error (Stinchcombe et al. 2008)

#Correlational selection

Another example of nonlinear selection is correlational selection, in which individuals with specific combinations of trait values show higher fitness (Phillips and Arnold 1989). This mode of selection is known as correlational selection and implies that selection on one trait depends on the value of another trait. Once again, we can use Lande and Arnold's regression approach to quantify correlational selection. To do that we add the cross-product terms $\gamma_{i,j}$ for each pair of traits to the regression model with the linear and the quadratic regression terms. For two traits z_i and z_j the model can be written as:

$w = \alpha + z_i\beta_i + z_j\beta_j + 1/2z_i^2\gamma_{i,i} + 1/2z_j^2\gamma_{j,j} + z_iz_j\gamma_{i,j} + \epsilon$

where $\gamma_{i,j}$ denotes the correlational selection gradient for traits z_i and z_j

To explore nonlinear selection, we will use a dataset from a study by Campbell et al. (2022) on *Ipomopsis*

Exercise 2: Selection by pollinators and seed predators on floral traits in two *Ipomopsis* species.

In this work by Dianne Campbell et al. (Campbell et al. 2022), they investigated linear and nonlinear phenotypic selection acting on a set of floral traits, including floral size-related traits, nectar, color and scent, in two species of *Ipomopsis* (pollinated by hummingbirds and hawk moths) and their hybrids during two different stages of the plant's life cycle which are: 1) pollination (i.e. seeds initiated) and 2) seed predation (proportion of fruits escaping fly predation).

Previous work on this system has shown that certain traits are under directional selection by pollinators and seed predators. Based on this knowledge, here they tested several predictions in which correlational selection may arise as the effect of combining directional selection by both pollinators and seed predators.

Here, we will focus on *I. aggregata* and three traits (floral width, color and nectar production) for which sample sizes are large enough to test for linear, quadratic and correlational selection through multivariate regression following Lande and Arnold's approach.

```
camp22= read.csv("Campbell_etal_22.csv")
names(camp22)
```

```
## [1] "Site"      "Year"      "Type"      "width"     "color"     "nectar"
## [7] "sepal"     "totpinene" "indoleN"   "seeds"     "seedsinit" "flyescape"
```

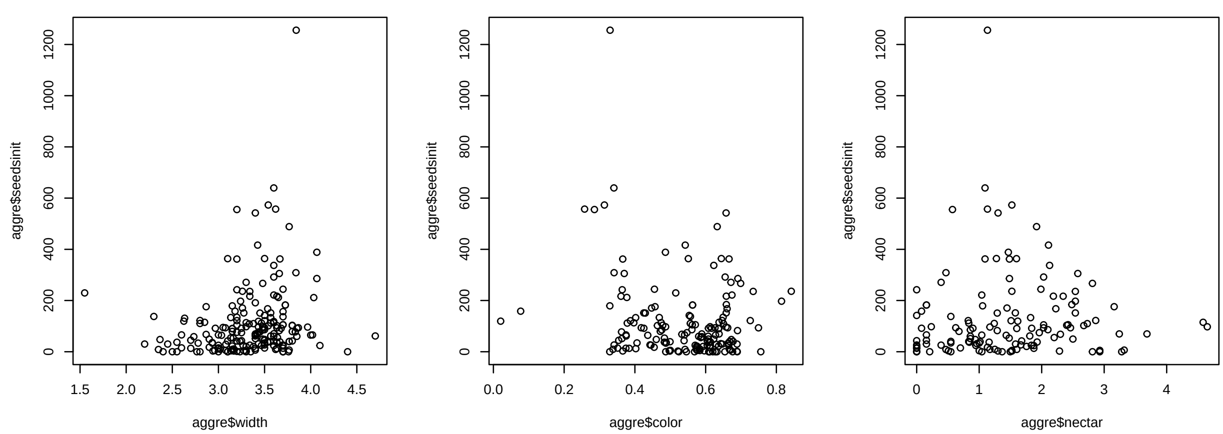
```
camp22$Site= as.factor(camp22$Site)
camp22$Type= as.factor(camp22$Type)
```

#Filter to include only I. aggregata sites

```
aggre= camp22 %>%
  filter(Site== "agg") %>%
dplyr::select(c(1:6, 11, 12))%>% #Filter to include petal width, color #and nectar
droplevels()
```

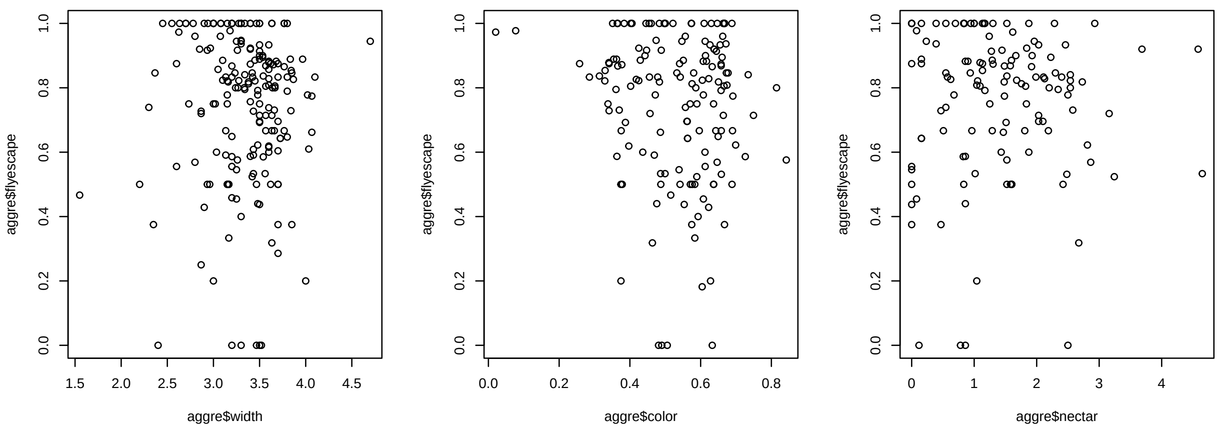
Explore the data: at pollination stage (seeds initiated)

```
par(mfrow= c(1,3))
plot(aggre$width, aggre$seedsinit)
plot(aggre$color, aggre$seedsinit)
plot(aggre$nectar, aggre$seedsinit)
```



Explore the data: at predation stage (seeds escaping fly predation)

```
par(mfrow= c(1,3))
plot(aggre$width, aggre$flyescape)
plot(aggre$color, aggre$flyescape)
plot(aggre$nectar, aggre$flyescape)
```



Estimate relative fitness at pollination and seed predation stages:

```r
aggre$relfit.pol = aggre$seedsinit/mean(aggre$seedsinit, na.rm = T)
aggre$relfit.pred = aggre$flyescape/mean(aggre$flyescape, na.rm = T)
```

#Variance-scaled univariate linear selection gradients at the pollination stage

```r
summary(lm(relfit.pol~ scale(width), data= aggre, na= na.exclude))$coef
```

```
##                Estimate Std. Error   t value     Pr(>|t|)
## (Intercept)   1.0157628 0.09402000 10.803690 1.097042e-21
## scale(width)  0.2722926 0.09519785  2.860281 4.686646e-03
```

```r
summary(lm(relfit.pol~ scale(color), data= aggre, na= na.exclude))$coef
```

```
##                 Estimate Std. Error   t value     Pr(>|t|)
## (Intercept)    1.0998237  0.1134841  9.691433 8.783222e-18
## scale(color) -0.2949649  0.1122830 -2.626977 9.446904e-03
```

```r
summary(lm(relfit.pol~ scale(nectar), data=aggre, na= na.exclude))$coef
```

```
##                  Estimate Std. Error   t value     Pr(>|t|)
## (Intercept)    1.21404288  0.1370054 8.8612749 4.676412e-15
## scale(nectar) 0.07696696  0.1374180 0.5600939 5.763647e-01
```

We observe positive directional selection to increase flower width and negative directional selection to decrease petal color (selection for pale-colored flowers)

Now when we accounting for selection acting on other traits: #Variance-scaled multivariate linear selection gradients at the pollination stage

```r
m = lm(relfit.pol~ scale(width) + scale(color)
+ scale(nectar), data= aggre, na= na.exclude)
summary(m)$coef
```

```
##                   Estimate Std. Error    t value     Pr(>|t|)
## (Intercept)    1.23422786  0.1316850  9.3725764 3.469012e-16
## scale(width)   0.48654477  0.1434723  3.3912115 9.279303e-04
## scale(color)  -0.30200110  0.1262586 -2.3919254 1.822624e-02
## scale(nectar)  0.05930215  0.1346535  0.4404055 6.603921e-01
```

We still detect linear selection through seeds initiated (i.e. pollination stage) acting on flower width (β_σ= 0.48) and petal color (β_σ= -0.30)

#Mean-scaled multivariate linear selection gradients at pollination stage

```
#Mean-scale the traits
aggre= aggre %>%
mutate(across(c(width, color, nectar),~ (.x-mean(.x, na.rm = T))/mean(.x, na.rm = T),
.names= "{col}_m"))


m = lm(relfit.pol~ width_m + color_m+ nectar_m, na= na.exclude, data= aggre)
summary(m)$coef
```

```
##                 Estimate Std. Error    t value     Pr(>|t|)
## (Intercept)  1.23422786  0.1316850  9.3725764 3.469012e-16
## width_m      3.86634780  1.1401081  3.3912115 9.279303e-04
## color_m     -1.25863303  0.5262008 -2.3919254 1.822624e-02
## nectar_m     0.08658552  0.1966041  0.4404055 6.603921e-01
```

Mean-standardized multivariate selection gradients also show strong directional selection on flower width (positive) and floral color (negative), with both β_μ's >1 (3.86 and -1.25) which are stronger than selection on fitness as a trait.

#Estimating multivariate nonlinear selection gradients (quadratic and interaction terms): We fit one model with: Linear terms: linear selection Quadratic terms: stabilizing and disruptive selection Interaction terms for each pair of traits: correlational selection

To add the quadratic terms to the model, we need to square the traits values:

```
#We first variance-scale the traits (here with "scale" function= variance-scaled traits)
#and then we square (^2) the traits:

aggre= aggre %>%
  mutate(across(c(width, color, nectar),  ~(scale(.)), .names= "{col}_s")) %>%
mutate(across(c(width_s, color_s, nectar_s),  ~ .x^2, .names= "{col}q"))
```

#Variance-scale multivariate nonlinear selection gradrients

```
m = lm(relfit.pol~ width_s + color_s+ nectar_s+
         width_sq + color_sq+ nectar_sq+
  width_s*color_s+ width_s*nectar_s+  color_s*nectar_s,
  na= na.exclude, data= aggre)
summary(m)$coef
```

```
##                      Estimate Std. Error    t value     Pr(>|t|)
## (Intercept)        1.00480974 0.20303951  4.9488384 2.440654e-06
```

```
## width_s           0.66668676 0.16838790  3.9592321 1.273946e-04
## color_s          -0.16445462 0.13661907 -1.2037457 2.310376e-01
## nectar_s          0.23777649 0.14370088  1.6546627 1.005846e-01
## width_sq          0.07139757 0.12127719  0.5887139 5.571506e-01
## color_sq          0.27747596 0.08799030  3.1534834 2.034931e-03
## nectar_sq        -0.15437757 0.09101317 -1.6962113 9.241720e-02
## width_s:color_s  -0.51241680 0.15674675 -3.2690745 1.405132e-03
## width_s:nectar_s -0.01008042 0.15178227 -0.0664137 9.471581e-01
## color_s:nectar_s -0.06011296 0.15145515 -0.3969027 6.921384e-01
```

There is Disruptive selection on floral color (color_sq$coeff= 0.27 >0) To estimate the quadratic selection gradient γ_i we double the quadratic regression coefficient and the standard error for flower color:

```
gamma_color =summary(m)$coef[6, 1]*2
SE= 0.091*2

gamma_color
```

```
## [1] 0.5549519
```

```
#gamma for floral color= 0.55 (0.18)
```

The model also indicates negative correlational selection ($\gamma_{i,j}$= -0.512) acting on flower width and color. This, selection favors plants with wide flowers when flowers are pale in color.

#Mean-scaled multivariate nonlinear selection gradients:

```
#Square the mean-scaled traits
aggre= aggre %>%
mutate(across(c(width_m, color_m, nectar_m),  ~ .x^2, .names= "{col}q"))

m = lm(relfit.pol~ width_m + color_m+ nectar_m+ width_mq +
         color_mq+ nectar_mq+ width_m*color_m+ width_m*nectar_m+
         color_m*nectar_m, na= na.exclude, data= aggre)
summary(m)$coef
```

```
##                     Estimate Std. Error    t value     Pr(>|t|)
## (Intercept)        1.0048097  0.2030395  4.9488384 2.440654e-06
## width_m            5.2978535  1.3381013  3.9592321 1.273946e-04
## color_m           -0.6853883  0.5693796 -1.2037457 2.310376e-01
## nectar_m           0.3471713  0.2098139  1.6546627 1.005846e-01
## width_mq           4.5085785  7.6583524  0.5887139 5.571506e-01
## color_mq           4.8195508  1.5283260  3.1534834 2.034931e-03
## nectar_mq         -0.3291046  0.1940233 -1.6962113 9.241720e-02
## width_m:color_m  -16.9703997  5.1911940 -3.2690745 1.405132e-03
## width_m:nectar_m  -0.1169585  1.7610594 -0.0664137 9.471581e-01
## color_m:nectar_m  -0.3657915  0.9216150 -0.3969027 6.921384e-01
```